

Q1 - 25 July - Shift 2

Let $x = \sin(2 \tan^{-1} \alpha)$ and $y = \sin\left(\frac{1}{2} \tan^{-1} \frac{4}{3}\right)$. If

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$S = \{\alpha \in \mathbb{R} : y^2 = 1 - x\}$, then $\sum_{\alpha \in S} 16\alpha^3$ is equal to

Q2 - 26 July - Shift 1

$\tan\left(2 \tan^{-1} \frac{1}{5} + \sec^{-1} \frac{\sqrt{5}}{2} + 2 \tan^{-1} \frac{1}{8}\right)$ is equal to:

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(A) 1

(B) 2

(C) $\frac{1}{4}$

(D) $\frac{5}{4}$

Q3 - 26 July - Shift 2

If $0 < x < \frac{1}{\sqrt{2}}$ and $\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta}$, then a value

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of $\sin\left(\frac{2\pi\alpha}{\alpha + \beta}\right)$ is

(A) $4\sqrt{(1-x^2)}(1-2x^2)$

(B) $4x\sqrt{(1-x^2)}(1-2x^2)$

(C) $2x\sqrt{(1-x^2)}(1-4x^2)$

(D) $4\sqrt{(1-x^2)}(1-4x^2)$

Q4 - 27 July - Shift 1

Questions

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For $k \in \mathbb{R}$, let the solutions of the equation

$$\cos\left(\sin^{-1}\left(x \cot\left(\tan^{-1}\left(\cos\left(\sin^{-1}x\right)\right)\right)\right)\right) = k, 0 < |x| < \frac{1}{\sqrt{2}}$$

be α and β , where the inverse trigonometric functions take only principal values. If the solutions of the equation $x^2 - bx - 5 = 0$ are

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} \text{ and } \frac{\alpha}{\beta}, \text{ then } \frac{b}{k^2} \text{ is equal to } \underline{\hspace{2cm}}.$$

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Q5 - 28 July - Shift 1

Considering the principal values of the inverse trigonometric functions, the sum of all the solutions of the equation $\cos^{-1}(x) - 2\sin^{-1}(x) = \cos^{-1}(2x)$ is equal to :

- (A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

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Answer Key

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Q1 (130)

Q2 (B)

Q3 (B)

Q4 (12)

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Q5 (A)

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Q1 (130)

(130)

Q2 (B)

$$\tan\left(2\left(\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8}\right) + \tan^{-1}\left(\frac{1}{2}\right)\right)$$

$$= \tan\left[2\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{2}\right)\right]$$

$$= 2$$

Q3 (B)

$$\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta} = k$$

$$\sin^{-1} x = k\alpha$$

$$\cos^{-1} x = k\beta$$

$$k = \frac{\pi}{2(\alpha + \beta)} \dots\dots(i)$$

$$\sin\left(\frac{2\pi\alpha}{\alpha + \beta}\right) = \sin(4\sin^{-1} x)$$

$$= 2\sin(2\sin^{-1} x) \cos(2\sin^{-1} x)$$

$$= 4x\sqrt{1-x^2}(1-2x^2)$$

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Q4 (12)

$$\cos(\sin^{-1} x) = \cos(\cos^{-1} \sqrt{1-x^2}) = \sqrt{1-x^2}$$

$$\cot(\tan^{-1} \sqrt{1-x^2}) = \cot \cot^{-1} \left(\sqrt{\frac{1}{1-x^2}} \right) = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \cos \left(\sin^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) \right) = \frac{\sqrt{1-2x^2}}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{\sqrt{1-2x^2}}{\sqrt{1-x^2}} = k$$

$$\Rightarrow 1-2x^2 = k^2(1-x^2)$$

$$\Rightarrow (k^2-2)x^2 = k^2-1$$

$$x^2 = \frac{k^2-1}{k^2-2}$$

$$\alpha = \sqrt{\frac{k^2-1}{k^2-2}} \Rightarrow \alpha^2 = \frac{k^2-1}{k^2-2}$$

$$\beta = \sqrt{\frac{k^2-1}{k^2-2}} \Rightarrow \beta^2 = \frac{k^2-1}{k^2-2}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = 2 \left(\frac{k^2-2}{k^2-1} \right) \& \frac{\alpha}{\beta} = -1$$

$$\text{Sum of roots} = \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{\alpha}{\beta} = b$$

$$\Rightarrow \frac{2(k^2-2)}{k^2-1} - 1 = b \dots\dots(1)$$

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Product of roots = $\left(\frac{1}{\alpha^2} + \frac{1}{\beta^2} \right) \frac{\alpha}{\beta} = -5$

Hints and Solutions

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$$\Rightarrow \frac{2(k^2 - 2)}{k^2 - 1}(-1) = -5$$

$$\Rightarrow 2k^2 - 4 = 5k^2 - 5$$

$$\Rightarrow 3k^2 = 1 \Rightarrow k^2 = \frac{1}{3} \dots \text{Put in (1)}$$

$$\Rightarrow b = \frac{2(k^2 - 2)}{k^2 - 1} - 1 = 5 - 1 = 4$$

$$\frac{b}{k^2} = \frac{4}{\frac{1}{3}} = 12$$

Q5 (A)

$$\cos^{-1} x = 2 \sin^{-1} x = \cos^{-1} 2x$$

$$\cos^{-1} x - 2 \left(\frac{\pi}{2} - \cos^{-1} x \right) = \cos^{-1} 2x$$

$$\cos^{-1} x - \pi + 2 \cos^{-1} x = \cos^{-1} 2x$$

$$3 \cos^{-1} x = \pi + \cos^{-1} 2x \dots (1)$$

$$\cos(3 \cos^{-1} x) = \cos(\pi + \cos^{-1} 2x)$$

$$4x^3 - 3x = -2x$$

$$4x^3 = x \Rightarrow x = 0, \pm \frac{1}{2}$$

All satisfy the original equation

$$\text{sum} = -\frac{1}{2} \text{ to } +\frac{1}{2} = 0$$

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